

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Worked Solutions

Quadratic Equations

Q001 Written

Solve $x^2 + 8 = 0$.**Solution**

1. $x^2 = -8$
2. $x = \pm\sqrt{-8} = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
3. Therefore the final answer is $x = \pm 2i\sqrt{2}$.

Final answer

$$x = \pm 2i\sqrt{2}$$

Q002 Written

Solve $x^2 + 4 = 0$.**Solution**

1. $x^2 = -4$
2. $x = \pm 2i$
3. Therefore the final answer is $x = \pm 2i$.

Final answer

$$x = \pm 2i$$

Worked Solutions

Quadratic Equations

Q003 Written

Solve $x^2 + 49 = 0$.

Solution

1. $x^2 = -49$
2. Therefore the final answer is $x = \pm 7i$.

Final answer

$$x = \pm 7i$$

Q004 Written

Solve $9x^2 - 4 = 0$.

Solution

1. $(3x - 2)(3x + 2) = 0$
2. Therefore the final answer is $x = \pm \frac{2}{3}$.

Final answer

$$x = \pm \frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q005 Written

Solve $x^2 - 12 = 0$.**Solution**

1. $x = \pm\sqrt{12}$
2. Therefore the final answer is $x = \pm 2\sqrt{3}$.

Final answer

$$x = \pm 2\sqrt{3}$$

Q006 Written

Solve $4x^2 + 3 = 0$.**Solution**

1. $x^2 = -3/4$
2. Therefore the final answer is $x = \pm \frac{i\sqrt{3}}{2}$.

Final answer

$$x = \pm \frac{i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2i\sqrt{2}$.

Correct option: A**Final answer**

$$x = \pm 2i\sqrt{2}$$

Q008 MCQ

Solve $x^2 + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2i$.

Correct option: A**Final answer**

$$x = \pm 2i$$

Worked Solutions

Quadratic Equations

Q009 MCQ

Solve $x^2 + 49 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 7i$.

Correct option: C**Final answer**

$$x = \pm 7i$$

Q010 MCQ

Solve $9x^2 - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm \frac{2}{3}$.

Correct option: C**Final answer**

$$x = \pm \frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2\sqrt{3}$.

Correct option: B**Final answer**

$$x = \pm 2\sqrt{3}$$

Q012 MCQ

Solve $4x^2 + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm \frac{i\sqrt{3}}{2}$.

Correct option: B**Final answer**

$$x = \pm \frac{i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q013 Written

Solve $(x - 3)^2 = -8$.**Solution**

1. $x - 3 = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
2. Therefore the final answer is $x = 3 \pm 2i\sqrt{2}$.

Final answer

$$x = 3 \pm 2i\sqrt{2}$$

Q014 Written

Solve $(x - 2)^2 = 5$.**Solution**

1. $x - 2 = \pm\sqrt{5}$
2. Therefore the final answer is $x = 2 \pm \sqrt{5}$.

Final answer

$$x = 2 \pm \sqrt{5}$$

Worked Solutions

Quadratic Equations

Q015 Written

Solve $(x + 6)^2 = -6$.**Solution**

1. $x + 6 = \pm i\sqrt{6}$
2. Therefore the final answer is $x = -6 \pm i\sqrt{6}$.

Final answer

$$x = -6 \pm i\sqrt{6}$$

Q016 Written

Solve $(x + 4)^2 = -18$.**Solution**

1. $x + 4 = \pm i\sqrt{18} = \pm 3i\sqrt{2}$
2. Therefore the final answer is $x = -4 \pm 3i\sqrt{2}$.

Final answer

$$x = -4 \pm 3i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q017 Written

Solve $(2x - 3)^2 = 7$.**Solution**

1. $2x - 3 = \pm\sqrt{7}$

2. Therefore the final answer is $x = \frac{3 \pm \sqrt{7}}{2}$.

Final answer

$$x = \frac{3 \pm \sqrt{7}}{2}$$

Q018 MCQ

Solve $(x - 3)^2 = -8$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = 3 \pm 2i\sqrt{2}$.**Correct option:** A**Final answer**

$$x = 3 \pm 2i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q019 MCQ

Solve $(x - 2)^2 = 5$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 2 \pm \sqrt{5}$.

Correct option: B**Final answer**

$$x = 2 \pm \sqrt{5}$$

Q020 MCQ

Solve $(x + 6)^2 = -6$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = -6 \pm i\sqrt{6}$.

Correct option: C**Final answer**

$$x = -6 \pm i\sqrt{6}$$

Worked Solutions

Quadratic Equations

Q021 MCQ

Solve $(x + 4)^2 = -18$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = -4 \pm 3i\sqrt{2}$.

Correct option: C**Final answer**

$$x = -4 \pm 3i\sqrt{2}$$

Q022 MCQ

Solve $(2x - 3)^2 = 7$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{3 \pm \sqrt{7}}{2}$.

Correct option: D**Final answer**

$$x = \frac{3 \pm \sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q023 Written

Solve $2x^2 - 7x - 9 = 0$.

Solution

1. $(x + 1)(2x - 9) = 0$
2. $x = -1$ or $x = 9/2$
3. Therefore the final answer is $x = -1, \frac{9}{2}$.

Final answer

$$x = -1, \frac{9}{2}$$

Q024 Written

Solve $2x^2 + 5x - 3 = 0$.

Solution

1. $(2x - 1)(x + 3) = 0$
2. Therefore the final answer is $x = \frac{1}{2}, -3$.

Final answer

$$x = \frac{1}{2}, -3$$

Worked Solutions

Quadratic Equations

Q025 Written

Solve $2x^2 + 7x = 0$.**Solution**

1. $x(2x + 7) = 0$
2. Therefore the final answer is $x = 0, -\frac{7}{2}$.

Final answer

$$x = 0, -\frac{7}{2}$$

Q026 Written

Solve $2x^2 + 5x - 12 = 0$.**Solution**

1. $(2x - 3)(x + 4) = 0$
2. Therefore the final answer is $x = \frac{3}{2}, -4$.

Final answer

$$x = \frac{3}{2}, -4$$

Worked Solutions

Quadratic Equations

Q027 **Written****Solve** $x^2 - 8x + 15 = 0$.**Solution**

1. $(x - 3)(x - 5) = 0$
2. Therefore the final answer is $x = 3, 5$.

Final answer $x = 3, 5$ Q028 **MCQ****Solve** $2x^2 - 7x - 9 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = -1, \frac{9}{2}$.

Correct option: A**Final answer** $x = -1, \frac{9}{2}$

Worked Solutions

Quadratic Equations

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{1}{2}, -3$.

Correct option: D**Final answer**

$$x = \frac{1}{2}, -3$$

Q030 MCQ

Solve $2x^2 + 7x = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 0, -\frac{7}{2}$.

Correct option: D**Final answer**

$$x = 0, -\frac{7}{2}$$

Worked Solutions

Quadratic Equations

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{3}{2}, -4$.

Correct option: A**Final answer**

$$x = \frac{3}{2}, -4$$

Q032 MCQ

Solve $x^2 - 8x + 15 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 3, 5$.

Correct option: A**Final answer**

$$x = 3, 5$$

Worked Solutions

Quadratic Equations

Q033 Written

Solve $3x^2 - 4x - 1 = 0$.**Solution**

1. $D = 16 + 12 = 28$
2. $x = (4 \pm 2\sqrt{7})/6 = (2 \pm \sqrt{7})/3$
3. Therefore the final answer is $x = \frac{2 \pm \sqrt{7}}{3}$.

Final answer

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Q034 MCQ

Solve $3x^2 - 4x - 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{2 \pm \sqrt{7}}{3}$.

Correct option: B**Final answer**

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Worked Solutions

Quadratic Equations

Q035 Written

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$.

Solution

1. $D = (2\sqrt{6})^2 - 4(2)(11) = -64$

2. $x = (-2\sqrt{6} \pm \sqrt{(-64)})/4 = (-\sqrt{6} \pm 4i)/2$

3. Therefore the final answer is $x = \frac{-\sqrt{6} \pm 4i}{2}$.

Final answer

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Q036 Written

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$.

Solution

1. $D = 12 - 40 = -28$

2. $x = (-2\sqrt{3} \pm i\sqrt{28})/4 = (-\sqrt{3} \pm i\sqrt{7})/2$

3. Therefore the final answer is $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q037 Written

Solve $x^2 + x + 1 = 0$.**Solution**

1. $D = 1 - 4 = -3$
2. $x = (-1 \pm i\sqrt{3})/2$
3. Therefore the final answer is $x = \frac{-1 \pm i\sqrt{3}}{2}$.

Final answer

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Q038 Written

Solve $3x^2 - 5x + 3 = 0$.**Solution**

1. $D = 25 - 36 = -11$
2. $x = (5 \pm i\sqrt{11})/6$
3. Therefore the final answer is $x = \frac{5 \pm i\sqrt{11}}{6}$.

Final answer

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Worked Solutions

Quadratic Equations

Q039 Written

Solve $x^2 - 2\sqrt{3}x + 5 = 0$.**Solution**

1. $D = 12 - 20 = -8$
2. $x = (2\sqrt{3} \pm i\sqrt{8})/2 = \sqrt{3} \pm i\sqrt{2}$
3. Therefore the final answer is $x = \sqrt{3} \pm i\sqrt{2}$.

Final answer

$$x = \sqrt{3} \pm i\sqrt{2}$$

Q040 Written

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$.**Solution**

1. $D = 8 - 12 = -4$
2. $x = (-2\sqrt{2} \pm 2i)/6 = (-\sqrt{2} \pm i)/3$
3. Therefore the final answer is $x = \frac{-\sqrt{2} \pm i}{3}$.

Final answer

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Worked Solutions

Quadratic Equations

Q041 Written

Solve $x^2 - x + 2 = 0$.**Solution**

1. $D = 1 - 8 = -7$
2. $x = (1 \pm i\sqrt{7})/2$
3. Therefore the final answer is $x = \frac{1 \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Q042 Written

Solve $3x^2 - \sqrt{5}x + 1 = 0$.**Solution**

1. $D = 5 - 12 = -7$
2. $x = (\sqrt{5} \pm i\sqrt{7})/6$
3. Therefore the final answer is $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$.

Final answer

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Worked Solutions

Quadratic Equations

Q043 **Written****Solve** $6x^2 - 12x + 15 = 0$.**Solution**

1. $D = 144 - 360 = -216$
2. $x = (12 \pm i6\sqrt{6})/12 = 1 \pm i\sqrt{6}/2$
3. Therefore the final answer is $x = 1 \pm \frac{i\sqrt{6}}{2}$.

Final answer

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Q044 **MCQ****Solve** $2x^2 + 2\sqrt{6}x + 11 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{-\sqrt{6} \pm 4i}{2}$.

Correct option: B**Final answer**

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Worked Solutions

Quadratic Equations

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$.**Correct option:** C**Final answer**

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = \frac{-1 \pm i\sqrt{3}}{2}$.**Correct option:** D**Final answer**

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{5 \pm i\sqrt{11}}{6}$.

Correct option: B**Final answer**

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Q048 MCQ

Solve $x^2 - 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \sqrt{3} \pm i\sqrt{2}$.

Correct option: D**Final answer**

$$x = \sqrt{3} \pm i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q049 MCQ

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = \frac{-\sqrt{2} \pm i}{3}$.**Correct option:** B**Final answer**

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Q050 MCQ

Solve $x^2 - x + 2 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = \frac{1 \pm i\sqrt{7}}{2}$.**Correct option:** C**Final answer**

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q051 MCQ

Solve $3x^2 - \sqrt{5}x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$.**Correct option:** C**Final answer**

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = 1 \pm \frac{i\sqrt{6}}{2}$.**Correct option:** B**Final answer**

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Worked Solutions

Quadratic Equations

Q053 Written

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.**Solution**

1. $D = 4^2 - 4(3)(7) = -68 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q054 Written

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.**Solution**

1. $D = (-m)^2 - 4(m^2 - 3m - 9) = -3m^2 + 12m + 36$
2. $D \geq 0 \rightarrow m^2 - 4m - 12 \leq 0 \rightarrow (m + 2)(m - 6) \leq 0$
3. Therefore the final answer is $-2 \leq m \leq 6$.

Final answer $-2 \leq m \leq 6$

Worked Solutions

Quadratic Equations

Q055 Written

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = k^2 - 4(3 - k) = (k - 2)(k + 6)$
2. Use $D > 0, D = 0, D < 0$ intervals
3. Therefore the final answer is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Written

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.

Solution

1. $D = 25 - 48 = -23 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q057 **Written****For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.****Solution**

1. $D = 49 + 32 = 81 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q058 **Written****For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.****Solution**

1. $D = 100 - 100 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q059 Written

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.**Solution**

1. $D = 25 - 12 = 13 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q060 Written

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.**Solution**

1. $D = 144 - 144 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q061 Written

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.**Solution**

1. $D = 64 - 80 = -16 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q062 Written

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.**Solution**

1. $D = 16 - 16 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q063 Written

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Solution

1. $D = 9 - 20 = -11 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q064 Written

Determine the nature of the solutions of $x^2 - x - 2 = 0$.

Solution

1. $D = 1 + 8 = 9 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-2 \leq m \leq 6$.

Correct option: D

Final answer

$$-2 \leq m \leq 6$$

Worked Solutions

Quadratic Equations

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Correct option: C

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: C

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: A

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q071 MCQ

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: C**Final answer**

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: C**Final answer**

One repeated real solution

Worked Solutions

Quadratic Equations

Q073 MCQ

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: B**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: A**Final answer**

One repeated real solution

Worked Solutions

Quadratic Equations

Q075 MCQ

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: A**Final answer**

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q077 Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Solution

1. $D = 4m^2 - 4(m + 2)$
2. $D < 0 \rightarrow m^2 - m - 2 < 0 \rightarrow (m - 2)(m + 1) < 0$
3. Therefore the final answer is $-1 < m < 2$.

Final answer

$$-1 < m < 2$$

Q078 Written

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 - 4(2a - 3) = (a - 2)(a - 6)$
2. Classify by sign of D
3. Therefore the final answer is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Final answer $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions

Quadratic Equations

Q079 Written

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution.

Solution

1. $D = (a + 1)^2 - 16 = 0$
2. $a + 1 = \pm 4$
3. Therefore the final answer is $a = 3$ or $a = -5$.

Final answer $a = 3$ or $a = -5$

Q080 Written

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions.

Solution

1. $D = (m + 6)^2 + 8m = m^2 + 20m + 36 = (m + 2)(m + 18)$
2. $D < 0$
3. Therefore the final answer is $-18 < m < -2$.

Final answer $-18 < m < -2$

Worked Solutions

Quadratic Equations

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Solution

1. $D = 4m^2 - 4(2m^2 - 3m) = 4m(3 - m)$
2. $D \geq 0$
3. Therefore the final answer is $0 \leq m \leq 3$.

Final answer

$$0 \leq m \leq 3$$

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-1 < m < 2$.

Correct option: C**Final answer**

$$-1 < m < 2$$

Worked Solutions

Quadratic Equations

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Correct option: C

Final answer

$a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions

Quadratic Equations

Q084 MCQ

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a = 3$ or $a = -5$.

Correct option: A**Final answer** $a = 3$ or $a = -5$

Q085 MCQ

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-18 < m < -2$.

Correct option: D**Final answer** $-18 < m < -2$

Worked Solutions

Quadratic Equations

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $0 \leq m \leq 3$.

Correct option: C

Final answer

$$0 \leq m \leq 3$$

Q087 Written

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.

Solution

1. Use $S = -b/a = 6/3 = 2$ and $P = c/a = -5/3$
2. Therefore the final answer is $\alpha + \beta = 2$, $\alpha\beta = -\frac{5}{3}$.

Final answer

$$\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$$

Worked Solutions

Quadratic Equations

Q088 Written

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.**Solution**

1. $S = -2/4 = -1/2; P = -7/4$
2. Therefore the final answer is $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$.

Final answer

$$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$$

Q089 Written

Find the sum and product of the roots of $x^2 + x + 2 = 0$.**Solution**

1. $S = -1; P = 2$
2. Therefore the final answer is $S = -1, \quad P = 2$.

Final answer

$$S = -1, \quad P = 2$$

Worked Solutions

Quadratic Equations

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.**Solution**

1. $S = 7/3; P = -2$
2. Therefore the final answer is $S = \frac{7}{3}, P = -2$.

Final answer

$$S = \frac{7}{3}, P = -2$$

Q091 MCQ

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$.

Correct option: C**Final answer**

$$\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$$

Worked Solutions

Quadratic Equations

Q092 MCQ

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $\alpha + \beta = -\frac{1}{2}$, $\alpha\beta = -\frac{7}{4}$.

Correct option: A**Final answer**

$$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$$

Q093 MCQ

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = -1$, $P = 2$.

Correct option: B**Final answer**

$$S = -1, \quad P = 2$$

Worked Solutions

Quadratic Equations

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = \frac{7}{3}$, $P = -2$.

Correct option: A**Final answer**

$$S = \frac{7}{3}, \quad P = -2$$

Q095 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.**Solution**

1. $S = 3, P = -3/2$
2. $S^2 - 2P = 9 + 3 = 12$
3. Therefore the final answer is 12.

Final answer

12

Worked Solutions

Quadratic Equations

Q096 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = -3/2$
2. $S^3 - 3PS = 27 + 27/2 = 81/2$
3. Therefore the final answer is $\frac{81}{2}$.

Final answer

$$\frac{81}{2}$$

Q097 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3/2$
2. $S^2 - 2P = 4 - 3 = 1$
3. Therefore the final answer is 1.

Final answer

$$1$$

Worked Solutions

Quadratic Equations

Q098 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3/2$
2. $S^3 - 3PS = 8 - 9 = -1$
3. Therefore the final answer is -1 .

Final answer -1

Q099 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3$
2. $S^2 - 2P = 4 - 6 = -2$
3. Therefore the final answer is -2 .

Final answer -2

Worked Solutions

Quadratic Equations

Q100 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3$
2. $S^3 - 3PS = 8 - 18 = -10$
3. Therefore the final answer is -10 .

Final answer

-10

Q101 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = -2, P = -1/3$
2. $S^2 - 2P = 4 + 2/3 = 14/3$
3. Therefore the final answer is $\frac{14}{3}$.

Final answer

$\frac{14}{3}$

Worked Solutions

Quadratic Equations

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = -2, P = -1/3$
2. $S^3 - 3PS = -8 - 2 = -10$
3. Therefore the final answer is -10 .

Final answer

-10

Q103 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 12.

Correct option: A

Final answer

12

Worked Solutions

Quadratic Equations

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{81}{2}$.

Correct option: B**Final answer**

$$\frac{81}{2}$$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 1.

Correct option: A**Final answer**

$$1$$

Worked Solutions

Quadratic Equations

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -1 .

Correct option: B**Final answer** -1

Q107 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -2 .

Correct option: D**Final answer** -2

Worked Solutions

Quadratic Equations

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -10 .

Correct option: B**Final answer** -10

Q109 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{14}{3}$.

Correct option: B**Final answer** $\frac{14}{3}$

Worked Solutions

Quadratic Equations

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -10 .

Correct option: D**Final answer** -10

Q111 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Solution

1. $2(\alpha + \beta)/(\alpha\beta) = 2(3)/(-3/2) = -4$
2. Therefore the final answer is -4 .

Final answer -4

Worked Solutions

Quadratic Equations

Q112 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Solution

1. $(\alpha^2 + \beta^2)/(\alpha\beta) = 1/(3/2) = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$$\frac{2}{3}$$

Q113 Written

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Solution

1. $S = 0; P = -3/5$
2. Therefore the final answer is $S = 0, P = -\frac{3}{5}$.

Final answer

$$S = 0, P = -\frac{3}{5}$$

Worked Solutions

Quadratic Equations

Q114 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$$\frac{2}{3}$$

Q115 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = (-2)/(-1/3) = 6$
2. Therefore the final answer is 6.

Final answer

$$6$$

Worked Solutions

Quadratic Equations

Q116 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -4 .

Correct option: C**Final answer** -4

Q117 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D**Final answer** $\frac{2}{3}$

Worked Solutions

Quadratic Equations

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = 0$, $P = -\frac{3}{5}$.

Correct option: D**Final answer**

$$S = 0, \quad P = -\frac{3}{5}$$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D**Final answer**

$$\frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q120 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 6.

Correct option: D**Final answer**

6

Q121 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Solution

1. $P + S + 1 = -3/2 + 3 + 1 = 5/2$
2. Therefore the final answer is $\frac{5}{2}$.

Final answer $\frac{5}{2}$

Worked Solutions

Quadratic Equations

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Solution

1. $P - S + 1 = 3/2 - 2 + 1 = 1/2$
2. Therefore the final answer is $\frac{1}{2}$.

Final answer

$$\frac{1}{2}$$

Q123 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Solution

1. $P + 2S + 4 = 3 + 4 + 4 = 11$
2. Therefore the final answer is 11.

Final answer

$$11$$

Worked Solutions

Quadratic Equations

Q124 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Solution

1. $P - 2S + 4 = -1/3 + 4 + 4 = 23/3$
2. Therefore the final answer is $\frac{23}{3}$.

Final answer

$$\frac{23}{3}$$

Q125 **MCQ**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{5}{2}$.

Correct option: A**Final answer**

$$\frac{5}{2}$$

Worked Solutions

Quadratic Equations

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{1}{2}$.

Correct option: C

Final answer

$$\frac{1}{2}$$

Q127 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 11.

Correct option: D

Final answer

$$11$$

Worked Solutions

Quadratic Equations

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{23}{3}$.

Correct option: D

Final answer

$$\frac{23}{3}$$

Q129 Written

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Solution

1. Let roots $\alpha, 3\alpha$. $4\alpha = 8 \Rightarrow \alpha = 2$.
2. $k = \alpha(3\alpha) = 6$
3. Therefore the final answer is $k = 6, \quad x = 2, 6$.

Final answer

$$k = 6, \quad x = 2, 6$$

Worked Solutions

Quadratic Equations

Q130 Written

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Solution

1. Roots $\alpha, 2\alpha$; $3\alpha = -6 \Rightarrow \alpha = -2$; $m = 2\alpha^2 = 8$
2. Therefore the final answer is $m = 8$, $x = -2, -4$.

Final answer

$$m = 8, \quad x = -2, -4$$

Q131 Written

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Solution

1. Roots $\alpha, 2\alpha$; $2\alpha^2 = 1/8 \Rightarrow \alpha = \pm 1/4$.
2. $m = 8(3\alpha) = \pm 6$.
3. Therefore the final answer is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Final answer

$$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2}) \text{ or } (-6, -\frac{1}{4}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k = 6$, $x = 2, 6$.

Correct option: A

Final answer

$$k = 6, \quad x = 2, 6$$

Q133 MCQ

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m = 8$, $x = -2, -4$.

Correct option: A

Final answer

$$m = 8, \quad x = -2, -4$$

Worked Solutions

Quadratic Equations

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Correct option: D

Final answer

$$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2}) \text{ or } (-6, -\frac{1}{4}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q135 Written

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots.

Solution

1. Let roots be α and $\alpha + 5$.
2. $\alpha(\alpha + 5) = -6 \Rightarrow \alpha = -2$ or -3 .
3. $S = 2\alpha + 5 = m - 1$ gives $m = 2$ or 0 ; obtain the paired roots.
4. Therefore the final answer is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Final answer

$$(m, x_1, x_2) = (2, -2, 3) \text{ or } (0, -3, 2)$$

Q136 Written

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots.

Solution

1. Let roots be α and $\alpha + 1$. $\alpha(\alpha + 1) = 2 \Rightarrow \alpha = 1$ or -2 .
2. $m + 1 = 2\alpha + 1 \Rightarrow m = 2$ or -4 .
3. Therefore the final answer is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Final answer

$$(m, x_1, x_2) = (2, 1, 2) \text{ or } (-4, -2, -1)$$

Worked Solutions

Quadratic Equations

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Solution

1. Roots $\alpha, \alpha + 4$; $2\alpha + 4 = -6 \rightarrow \alpha = -5$; $m = \alpha(\alpha + 4) = 5$
2. Therefore the final answer is $m = 5$, $x = -5, -1$.

Final answer

$$m = 5, \quad x = -5, -1$$

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Solution

1. Roots $\alpha, \alpha + 1$; $\alpha(\alpha + 1) = 3/4 \rightarrow \alpha = 1/2 \text{ or } -3/2$.
2. $-m/4 = 2\alpha + 1 \rightarrow m = -8 \text{ or } 8$.
3. Therefore the final answer is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Final answer

$$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2}) \text{ or } (8, -\frac{3}{2}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q139 MCQ

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Correct option: D

Final answer

$$(m, x_1, x_2) = (2, -2, 3) \text{ or } (0, -3, 2)$$

Worked Solutions

Quadratic Equations

Q140 MCQ

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Correct option: A

Final answer

$(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Worked Solutions

Quadratic Equations

Q141 MCQ

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m = 5$, $x = -5, -1$.

Correct option: B

Final answer

$$m = 5, \quad x = -5, -1$$

Worked Solutions

Quadratic Equations

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Correct option: B

Final answer

$$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2}) \text{ or } (8, -\frac{3}{2}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q143 Written

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots.

Solution

- Let roots $2t, 3t$; $5t = 2 \rightarrow t = 2/5$; $m = 6t^2 = 24/25$
- Therefore the final answer is $m = \frac{24}{25}$, $x = \frac{4}{5}, \frac{6}{5}$.

Final answer

$$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$$

Q144 MCQ

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots Select the correct answer:

Solution

- Apply the same quadratic method as the source demand.
- The correct value is $m = \frac{24}{25}$, $x = \frac{4}{5}, \frac{6}{5}$.

Correct option: A

Final answer

$$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$$

Worked Solutions

Quadratic Equations

Q145 Written

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Solution

1. Roots α, α^2 ; $\alpha + \alpha^2 = 6 \Rightarrow \alpha = 2$ or -3 .
2. $k = \alpha^3 = 8$ or -27 .
3. Therefore the final answer is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Final answer

$$(k, x_1, x_2) = (8, 2, 4) \text{ or } (-27, -3, 9)$$

Worked Solutions

Quadratic Equations

Q146 MCQ

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Correct option: B

Final answer

$$(k, x_1, x_2) = (8, 2, 4) \text{ or } (-27, -3, 9)$$

Worked Solutions

Quadratic Equations

Q147 Written

Factorise $2x^2 - 3x + 2$ over the complex numbers.

Solution

1. $D = 9 - 16 = -7$; roots $(3 \pm i\sqrt{7})/4$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Q148 Written

Factorise $x^2 + 2x + 5$ over the complex numbers.

Solution

1. Roots $-1 \pm 2i$
2. Therefore the final answer is $(x + 1 - 2i)(x + 1 + 2i)$.

Final answer

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions

Quadratic Equations

Q149 Written

Factorise $2x^2 - 6x + 1$ over the complex numbers.

Solution

1. $D = 28$; roots $(3 \pm \sqrt{7})/2$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$.

Final answer

$$2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$$

Q150 Written

Factorise $x^2 + 4x + 5$ over the complex numbers.

Solution

1. $D = 16 - 20 = -4$; roots $-2 \pm i$
2. Therefore the final answer is $(x + 2 - i)(x + 2 + i)$.

Final answer

$$(x + 2 - i)(x + 2 + i)$$

Worked Solutions

Quadratic Equations

Q151 **Written****Factorise $x^2 + 4x + 2$ over the complex numbers.****Solution**

1. *Roots* $-2 \pm \sqrt{2}$
2. Therefore the final answer is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Final answer

$$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

Q152 **Written****Factorise $x^2 - 4x + 7$ over the complex numbers.****Solution**

1. $D = 16 - 28 = -12$; *roots* $2 \pm i\sqrt{3}$
2. Therefore the final answer is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Final answer

$$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$$

Worked Solutions

Quadratic Equations

Q153 Written

Factorise $2x^2 - x + 1$ over the complex numbers.**Solution**

1. $D = 1 - 8 = -7$; roots $(1 \pm i\sqrt{7})/4$
2. Therefore the final answer is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Q154 Written

Factorise $3x^2 - 5x + 3$ over the complex numbers.**Solution**

1. $D = 25 - 36 = -11$; roots $(5 \pm i\sqrt{11})/6$
2. Therefore the final answer is $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$.

Final answer

$$3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$$

Worked Solutions

Quadratic Equations

Q155 **Written****Factorise $2x^2 - 2x - 1$ over the complex numbers.****Solution**

1. $D = 4 + 8 = 12$; roots $(1 \pm \sqrt{3})/2$

2. Therefore the final answer is $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$.

Final answer

$$2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q156 MCQ

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Correct option: C**Final answer**

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Worked Solutions

Quadratic Equations

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 1 - 2i)(x + 1 + 2i)$.

Correct option: A**Final answer**

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions

Quadratic Equations

Q158 MCQ

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$.

Correct option: D**Final answer**

$$2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 2 - i)(x + 2 + i)$.

Correct option: D**Final answer**

$$(x + 2 - i)(x + 2 + i)$$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Correct option: B**Final answer**

$$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Correct option: A**Final answer**

$$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$$

Worked Solutions

Quadratic Equations

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.**Correct option: C****Final answer**

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Worked Solutions

Quadratic Equations

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$.

Correct option: A**Final answer**

$$3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$$

Worked Solutions

Quadratic Equations

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$.

Correct option: B**Final answer**

$$2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q165 Written

Factorise $x^4 + 2x^2 - 3$ over the complex numbers.**Solution**

1. $x^4 + 2x^2 - 3 = (x^2 - 1)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$$(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$$

Q166 Written

Factorise $x^4 - x^2 - 6$ over the complex numbers.**Solution**

1. Let $y = x^2$: $y^2 - y - 6 = (y - 3)(y + 2)$
2. Factor $x^2 - 3$ and $x^2 + 2$
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$.

Final answer

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$$

Worked Solutions

Quadratic Equations

Q167 Written

Factorise $x^4 - 9$ over the complex numbers.

Solution

1. $x^4 - 9 = (x^2 - 3)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$$

Q168 Written

Factorise $x^4 - 3x^2 + 2$ over the complex numbers.

Solution

1. Let $y = x^2$: $y^2 - 3y + 2 = (y - 1)(y - 2)$
2. Factor $x^2 - 1$ and $x^2 - 2$
3. Therefore the final answer is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Final answer

$$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q169 MCQ

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: B**Final answer**

$$(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$$

Q170 MCQ

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$.

Correct option: B**Final answer**

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$$

Worked Solutions

Quadratic Equations

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: D**Final answer**

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$$

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Correct option: C**Final answer**

$$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q173 Written

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Solution

1. $S = 1, P = 1/2$
2. $x^2 - x + 1/2 = 0$; multiply by 2
3. Therefore the final answer is $2x^2 - 2x + 1 = 0$.

Final answer

$$2x^2 - 2x + 1 = 0$$

Q174 Written

Find two numbers whose sum is -1 and product is 2 .

Solution

1. They are roots of $x^2 + x + 2 = 0$
2. $D = -7$
3. Therefore the final answer is $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$.

Final answer

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q175 Written

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Solution

1. $S = 6, P = 13$
2. $x^2 - Sx + P = 0$
3. Therefore the final answer is $x^2 - 6x + 13 = 0$.

Final answer

$$x^2 - 6x + 13 = 0$$

Q176 Written

Find two numbers whose sum and product are both 1.

Solution

1. Roots of $x^2 - x + 1 = 0$
2. $D = -3$
3. Therefore the final answer is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.

Final answer

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q177 **Written****Create a quadratic equation with roots -1 and 2 .****Solution**

1. $S = 1, P = -2$
2. Therefore the final answer is $x^2 - x - 2 = 0$.

Final answer

$$x^2 - x - 2 = 0$$

Q178 **Written****Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.****Solution**

1. $S = 2, P = 17$
2. Therefore the final answer is $x^2 - 2x + 17 = 0$.

Final answer

$$x^2 - 2x + 17 = 0$$

Worked Solutions

Quadratic Equations

Q179 Written

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Solution

1. $S = 10/3, P = 2$
2. $x^2 - (10/3)x + 2 = 0$; multiply by 3
3. Therefore the final answer is $3x^2 - 10x + 6 = 0$.

Final answer

$$3x^2 - 10x + 6 = 0$$

Q180 Written

Find two numbers whose sum is 2 and product is 5.

Solution

1. Roots of $x^2 - 2x + 5 = 0$
2. $D = -16$
3. Therefore the final answer is $1 + 2i, 1 - 2i$.

Final answer

$$1 + 2i, 1 - 2i$$

Worked Solutions

Quadratic Equations

Q181 **Written****Find two numbers whose sum is 4 and product is 2.****Solution**

1. Roots of $x^2 - 4x + 2 = 0$
2. Therefore the final answer is $2 + \sqrt{2}$, $2 - \sqrt{2}$.

Final answer

$$2 + \sqrt{2}, 2 - \sqrt{2}$$

Q182 **MCQ****Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots Select the correct answer:****Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $2x^2 - 2x + 1 = 0$.

Correct option: B**Final answer**

$$2x^2 - 2x + 1 = 0$$

Worked Solutions

Quadratic Equations

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$.**Correct option: B****Final answer**

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Q184 MCQ

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x^2 - 6x + 13 = 0$.**Correct option: B****Final answer**

$$x^2 - 6x + 13 = 0$$

Worked Solutions

Quadratic Equations

Q185 MCQ

Find two numbers whose sum and product are both 1 Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.**Correct option:** A**Final answer**

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Q186 MCQ

Create a quadratic equation with roots -1 and 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.

2. The correct value is $x^2 - x - 2 = 0$.**Correct option:** A**Final answer**

$$x^2 - x - 2 = 0$$

Worked Solutions

Quadratic Equations

Q187 MCQ

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 2x + 17 = 0$.

Correct option: D

Final answer

$$x^2 - 2x + 17 = 0$$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $3x^2 - 10x + 6 = 0$.

Correct option: A

Final answer

$$3x^2 - 10x + 6 = 0$$

Worked Solutions

Quadratic Equations

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $1 + 2i$, $1 - 2i$.

Correct option: B**Final answer** $1 + 2i$, $1 - 2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the source demand.
2. The correct value is $2 + \sqrt{2}$, $2 - \sqrt{2}$.

Correct option: B**Final answer** $2 + \sqrt{2}$, $2 - \sqrt{2}$

Worked Solutions

Quadratic Equations

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Solution

1. $S = -2, P = -4$
2. $\text{Newsum} = S + 4 = 2; \text{newproduct} = P + 2S + 4 = -4$
3. $\text{Equation } x^2 - 2x - 4 = 0$
4. Therefore the final answer is $x^2 - 2x - 4 = 0$.

Final answer

$$x^2 - 2x - 4 = 0$$

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 2x - 4 = 0$.

Correct option: C

Final answer

$$x^2 - 2x - 4 = 0$$

Worked Solutions

Quadratic Equations

Q193 Written

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 .

Solution

1. $S = -1, P = -3$
2. $\text{Newsum} S^2 - 2P = 1 + 6 = 7$; $\text{product} P^2 = 9$
3. Therefore the final answer is $x^2 - 7x + 9 = 0$.

Final answer

$$x^2 - 7x + 9 = 0$$

Q194 Written

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 .

Solution

1. $S = 2, P = 7$
2. $\text{Newsum} = S^2 - 2P = -10$; $\text{product} = P^2 = 49$
3. Therefore the final answer is $x^2 + 10x + 49 = 0$.

Final answer

$$x^2 + 10x + 49 = 0$$

Worked Solutions

Quadratic Equations

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 7x + 9 = 0$.

Correct option: D**Final answer**

$$x^2 - 7x + 9 = 0$$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 + 10x + 49 = 0$.

Correct option: C**Final answer**

$$x^2 + 10x + 49 = 0$$

Worked Solutions

Quadratic Equations

Q197 Written

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Solution

1. $S = -3/2, P = 2$
2. $\text{Newsum} = S/P = -3/4$; $\text{product} = 1/P = 1/2$
3. $x^2 + (3/4)x + 1/2 = 0$; multiply by 4
4. Therefore the final answer is $4x^2 + 3x + 2 = 0$.

Final answer

$$4x^2 + 3x + 2 = 0$$

Q198 MCQ

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $4x^2 + 3x + 2 = 0$.

Correct option: A

Final answer

$$4x^2 + 3x + 2 = 0$$

Worked Solutions

Quadratic Equations

Q199 Written

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Solution

1. $S = -2, P = 3$
2. $\text{Newsum} = 4S = -8$
3. $\text{Newproduct} = 3(\alpha^2 + \beta^2) + 10P = 3(S^2 - 2P) + 10P = 24$
4. Therefore the final answer is $x^2 + 8x + 24 = 0$.

Final answer

$$x^2 + 8x + 24 = 0$$

Q200 MCQ

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 + 8x + 24 = 0$.

Correct option: D

Final answer

$$x^2 + 8x + 24 = 0$$

Worked Solutions

Quadratic Equations

Q201 Written

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m .

Solution

1. $D > 0 \rightarrow (m - 2)(m + 1) > 0 \rightarrow m < -1 \text{ or } m > 2$
2. $S = -2m > 0 \rightarrow m < 0$
3. $P = m + 2 > 0 \rightarrow m > -2$
4. Intersection : $-2 < m < -1$
5. Therefore the final answer is $-2 < m < -1$.

Final answer

$$-2 < m < -1$$

Worked Solutions

Quadratic Equations

Q202 Written

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m .

Solution

1. $D = 4(4 - m)(m + 1) > 0 \rightarrow -1 < m < 4$
2. $S = 2(1 - m) > 0 \rightarrow m < 1$
3. $P = (2m + 1)(m - 3) > 0 \rightarrow m < -1/2 \text{ or } m > 3$
4. Intersection : $-1 < m < -1/2$
5. Therefore the final answer is $-1 < m < -\frac{1}{2}$.

Final answer

$$-1 < m < -\frac{1}{2}$$

Worked Solutions

Quadratic Equations

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-2 < m < -1$.

Correct option: A**Final answer**

$$-2 < m < -1$$

Q204 MCQ

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-1 < m < -\frac{1}{2}$.

Correct option: B**Final answer**

$$-1 < m < -\frac{1}{2}$$

Worked Solutions

Quadratic Equations

Q205 **Written**

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m .

Solution

1. $D > 0 : 4m^2 - 4(m + 6) > 0 \rightarrow (m - 3)(m + 2) > 0$
2. $S = 2m < 0 \rightarrow m < 0$
3. $P = m + 6 > 0 \rightarrow m > -6$
4. *Intersection* : $-6 < m < -2$
5. Therefore the final answer is $-6 < m < -2$.

Final answer

$$-6 < m < -2$$

Worked Solutions

Quadratic Equations

Q206 **Written**

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k .

Solution

1. $D = (k - 10)(k + 2) > 0 \rightarrow k < -2 \text{ or } k > 10$
2. $S = -k < 0 \rightarrow k > 0$
3. $P = 2k + 5 > 0 \rightarrow k > -5/2$
4. *Intersection* $k > 10$
5. Therefore the final answer is $k > 10$.

Final answer $k > 10$

Worked Solutions

Quadratic Equations

Q207 MCQ

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-6 < m < -2$.

Correct option: C**Final answer**

$$-6 < m < -2$$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k > 10$.

Correct option: B**Final answer**

$$k > 10$$

Worked Solutions

Quadratic Equations

Q209 Written

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m .

Solution

1. Opposite signs if $P = -m - 4 < 0$
2. Therefore the final answer is $m > -4$.

Final answer

$$m > -4$$

Q210 Written

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k .

Solution

1. $P = -k - 3 < 0$
2. Therefore the final answer is $k > -3$.

Final answer

$$k > -3$$

Worked Solutions

Quadratic Equations

Q211 MCQ

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m > -4$.

Correct option: A**Final answer**

$$m > -4$$

Q212 MCQ

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k > -3$.

Correct option: D**Final answer**

$$k > -3$$

Worked Solutions

Quadratic Equations

Q213 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 + 4a = a(a + 4)$
2. Classify sign intervals at -4 and 0
3. Therefore the final answer is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions

Quadratic Equations

Q214 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = (k + 2)^2 - 4k^2 = -3k^2 + 4k + 4 = -3(k - 2)(k + 2/3)$

2. Classify D sign

3. Therefore the final answer is $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary.

Final answer

$-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q215 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = 4(a - 1)^2 - 4(a^2 - 3a + 4) = 4(a - 3)$
2. Classify by sign of $a - 3$
3. Therefore the final answer is $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary.

Final answer

$a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Worked Solutions

Quadratic Equations

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Solution

1. $\alpha\beta(\alpha + \beta) = PS = (3/2)(3) = 9/2$
2. Therefore the final answer is $\frac{9}{2}$.

Final answer

$$\frac{9}{2}$$

Worked Solutions

Quadratic Equations

Q217 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Solution

1. $S^2 - 4P = 9 - 6 = 3$
2. Therefore the final answer is 3.

Final answer

3

Worked Solutions

Quadratic Equations

Q218 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = 3/2$
2. $S^3 - 3PS = 27 - 27/2 = 27/2$
3. Therefore the final answer is $\frac{27}{2}$.

Final answer

$$\frac{27}{2}$$

Worked Solutions

Quadratic Equations

Q219 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Solution

1. $(\alpha^3 + \beta^3)/(\alpha\beta) = (27/2)/(3/2) = 9$
2. Therefore the final answer is 9.

Final answer

9

Worked Solutions

Quadratic Equations

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Solution

1. *Subtract equations* : $(a - b)(x - c) = 0$, so *common root* $r = c$.
2. *Substitute* c : $c(c + a + b) = 0$; $c \neq 0 \rightarrow a + b + c = 0$.
3. *The other roots are* $bc/c = b$ and $ca/c = a$. *Their equation is* $x^2 - (a + b)x + ab = x^2 + cx + ab$.
4. Therefore the final answer is Common root c ; other roots a and b .

Final answer

Common root c ; other roots a and b .

Worked Solutions

Quadratic Equations

Q221 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Solution

1. Let $y = x^2$: $6y^2 - 7y - 3 = (2y - 3)(3y + 1)$
2. $x^2 = 3/2$ or $-1/3$; hence $x = \pm\sqrt{6}/2$ or $\pm i\sqrt{3}/3$
3. For real displacement x , only $\pm\sqrt{6}/2$ are admissible
4. Therefore the final answer is $x = \pm\frac{\sqrt{6}}{2}$, $x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$.

Final answer

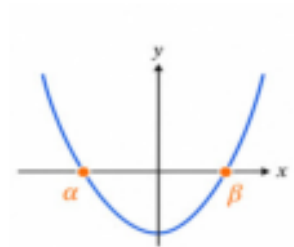
$$x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}; \text{ real: } \pm\sqrt{6}/2$$

Worked Solutions

Quadratic Equations

Q222 Challenge

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Solution

1. $\text{Newsum} = S/P; \text{newproduct} = 1/P$
2. Equation $x^2 - (S/P)x + 1/P = 0$; multiply by P
3. Therefore the final answer is For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$.

Final answer

For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$

Worked Solutions

Quadratic Equations

Q223 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Solution

1. $D = (m + 1)^2 - 4(m + 4) > 0 \rightarrow m < -3 \text{ or } m > 5$
2. $S = -(m + 1) > 0 \rightarrow m < -1$
3. $P = m + 4 > 0 \rightarrow m > -4$
4. Intersection $(-4, -3)$
5. Therefore the final answer is $-4 < m < -3$.

Final answer

$$-4 < m < -3$$

Worked Solutions

Quadratic Equations

Q224 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Solution

1. $D > 0 \rightarrow m < -3$ or $m > 5$
2. $S < 0 \rightarrow m > -1$
3. $P > 0 \rightarrow m > -4$
4. *Intersection* $m > 5$
5. Therefore the final answer is $m > 5$.

Final answer

$$m > 5$$

Worked Solutions

Quadratic Equations

Q225 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Solution

1. Opposite signs if $P = m + 4 < 0$
2. Therefore the final answer is $m < -4$.

Final answer

$$m < -4$$

Worked Solutions

Quadratic Equations

Q226 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Solution

1. Set $R = 0$. $D = 12 - 24 = -12$.
2. $x = (-2\sqrt{3} \pm i2\sqrt{3})/6 = (-\sqrt{3} \pm i\sqrt{3})/3$.
3. Because x is real, the complex pair is not admissible; no real equilibrium exists.
4. Therefore the final answer is $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}; \text{ no real equilibrium}$$

Worked Solutions

Quadratic Equations

Q227 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Correct option: B

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions

Quadratic Equations

Q228 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary.

Correct option: B

Final answer

$-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q229 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary.

Correct option: B

Final answer

$a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Worked Solutions

Quadratic Equations

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{9}{2}$.

Correct option: B

Final answer

$$\frac{9}{2}$$

Worked Solutions

Quadratic Equations

Q231 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 3.

Correct option: D

Final answer

3

Worked Solutions

Quadratic Equations

Q232 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{27}{2}$.

Correct option: B

Final answer

$$\frac{27}{2}$$

Worked Solutions

Quadratic Equations

Q233 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 9.

Correct option: A

Final answer

9

Worked Solutions

Quadratic Equations

Q234 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Common root c ; other roots a and b .

Correct option: C

Final answer

Common root c ; other roots a and b .

Worked Solutions

Quadratic Equations

Q235 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$.

Correct option: A

Final answer

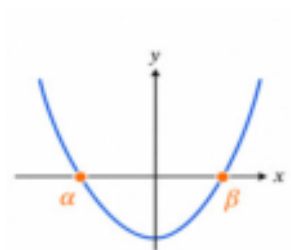
$$x = \pm \frac{\sqrt{6}}{2}, \quad x = \pm \frac{i\sqrt{3}}{3}; \text{ real: } \pm \sqrt{6}/2$$

Worked Solutions

Quadratic Equations

Q236 Challenge

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:



Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$.

Correct option: A

Final answer

For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$

Worked Solutions

Quadratic Equations

Q237 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-4 < m < -3$.

Correct option: C

Final answer

$$-4 < m < -3$$

Worked Solutions

Quadratic Equations

Q238 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m > 5$.

Correct option: D

Final answer

$$m > 5$$

Worked Solutions

Quadratic Equations

Q239 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m < -4$.

Correct option: D

Final answer

$$m < -4$$

Worked Solutions

Quadratic Equations

Q240 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result. Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Correct option: D

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}; \text{ no real equilibrium}$$

Worked Solutions

Quadratic Equations

Quiz MCQ 1 Concept Quiz · MCQ

Solve $x^2 - 5x + 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 2, 3.

Correct option: B

Final answer

2, 3

Worked Solutions

Quadratic Equations

Quiz MCQ 2 Concept Quiz · MCQ

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct non-real complex roots.

Correct option: A

Final answer

Two distinct non-real complex roots

Worked Solutions

Quadratic Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

If α, β are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 7.

Correct option: C

Final answer

7

Quiz MCQ 4 **Concept Quiz · MCQ**

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 4x + 5 = 0$.

Correct option: A

Final answer

$x^2 - 4x + 5 = 0$

Worked Solutions

Quadratic Equations

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $2x^2 + 3x - 2 = 0$.**Solution**

1. Apply the relevant quadratic identity.
2. Final answer: $x = \frac{1}{2}, -2$

Final answer

$$x = \frac{1}{2}, -2$$

Quiz WRITTEN 6 Concept Quiz · Written**Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.****Solution**

1. Apply the relevant quadratic identity.
2. Final answer: $-1 < m < 2$

Final answer

$$-1 < m < 2$$

Worked Solutions

Quadratic Equations

Quiz WRITTEN 7 Concept Quiz · Written

If α, β are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: 28

Final answer

28

Quiz WRITTEN 8 Concept Quiz · Written

If α, β are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2 .

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $x^2 + 6x + 25 = 0$

Final answer

$x^2 + 6x + 25 = 0$

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$

Q002 $x = \pm 2i$

Q003 $x = \pm 7i$

Q004 $x = \pm \frac{2}{3}$

Q005 $x = \pm 2\sqrt{3}$

Q006 $x = \pm \frac{i\sqrt{3}}{2}$

Q007 A

Q008 A

Q009 C

Q010 C

Q011 B

Q012 B

Q013 $x = 3 \pm 2i\sqrt{2}$

Q014 $x = 2 \pm \sqrt{5}$

Q015 $x = -6 \pm i\sqrt{6}$

Q016 $x = -4 \pm 3i\sqrt{2}$

Q017 $x = \frac{3 \pm \sqrt{7}}{2}$

Q018 A

Q019 B

Q020 C

Q021 C

Q022 D

Q023 $x = -1, \frac{9}{2}$

Q024 $x = \frac{1}{2}, -3$

Q025 $x = 0, -\frac{7}{2}$

Q026 $x = \frac{3}{2}, -4$

Q027 $x = 3, 5$

Q028 A

Q029 D

Q030 D

Q031 A

Q032 A

Q033 $x = \frac{2 \pm \sqrt{7}}{3}$

Q034 B

Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$

Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$

Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$

Q039 $x = \sqrt{3} \pm i\sqrt{2}$

Q040 $x = \frac{-\sqrt{2} \pm i}{3}$

Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$

Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$

Q044 B

Q045 C

Q046 D

Q047 B

Q048 D

Q049 B

Q050 C

Q051 C

Q052 B

Q053 Two distinct imaginary (non-real complex) solutions

Q054 $-2 \leq m \leq 6$

Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Two distinct imaginary (non-real complex) solutions

Q057 Two distinct real solutions

Q058 One repeated real solution

Q059 Two distinct real solutions

Q060 One repeated real solution

Q061 Two distinct imaginary (non-real complex) solutions

Q062 One repeated real solution

Q063 Two distinct imaginary (non-real complex) solutions

Q064 Two distinct real solutions

Q065 A

Q066 D

Q067 C

Q068 C

Q069 A

Q070 A

Q071 C

Q072 C

Q073 B

Q074 A

Q075 A

Q076 A

Q077 $-1 < m < 2$

Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Q079 $a = 3$ or $a = -5$

Q080 $-18 < m < -2$

Q081 $0 \leq m \leq 3$

Q082 C

Q083 C

Q084 A

Q085 D

Q086 C

Q087 $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$

Q088 $\alpha + \beta = -\frac{1}{2}, \alpha\beta = -\frac{7}{4}$

Q089 $S = -1, P = 2$

Q090 $S = \frac{7}{3}, P = -2$

Q091 C

Q092 A

Q093 B

Q094 A

Q095 12

Q096 $\frac{81}{2}$
 Q097 1
 Q098 -1
 Q099 -2
 Q100 -10
 Q101 $\frac{14}{3}$
 Q102 -10
 Q103 A
 Q104 B
 Q105 A
 Q106 B
 Q107 D
 Q108 B
 Q109 B
 Q110 D
 Q111 -4
 Q112 $\frac{2}{3}$
 Q113 $S = 0, \quad P = -\frac{3}{5}$
 Q114 $\frac{2}{3}$
 Q115 6
 Q116 C
 Q117 D
 Q118 D
 Q119 D
 Q120 D
 Q121 $\frac{5}{2}$
 Q122 $\frac{1}{2}$
 Q123 11
 Q124 $\frac{23}{3}$
 Q125 A

Q126 C
 Q127 D
 Q128 D
 Q129 $k = 6, \quad x = 2, 6$
 Q130 $m = 8, \quad x = -2, -4$
 Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$
 Q132 A
 Q133 A
 Q134 D
 Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$
 Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$
 Q137 $m = 5, \quad x = -5, -1$
 Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$
 Q139 D
 Q140 A
 Q141 B
 Q142 B
 Q143 $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$
 Q144 A
 Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$
 Q146 B
 Q147 $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$
 Q148 $(x+1-2i)(x+1+2i)$
 Q149 $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$
 Q150 $(x+2-i)(x+2+i)$
 Q151 $(x+2-\sqrt{2})(x+2+\sqrt{2})$
 Q152 $(x-2-i\sqrt{3})(x-2+i\sqrt{3})$
 Q153 $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$
 Q154 $3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$
 Q156 C
 Q157 A
 Q158 D
 Q159 D
 Q160 B
 Q161 A
 Q162 C
 Q163 A
 Q164 B
 Q165 $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$
 Q166 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$
 Q167 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$
 Q168 $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$
 Q169 B
 Q170 B
 Q171 D
 Q172 C
 Q173 $2x^2 - 2x + 1 = 0$
 Q174 $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$
 Q175 $x^2 - 6x + 13 = 0$
 Q176 $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
 Q177 $x^2 - x - 2 = 0$
 Q178 $x^2 - 2x + 17 = 0$
 Q179 $3x^2 - 10x + 6 = 0$
 Q180 $1+2i, 1-2i$
 Q181 $2+\sqrt{2}, 2-\sqrt{2}$
 Q182 B
 Q183 B
 Q184 B

Q185 A
 Q186 A
 Q187 D
 Q188 A
 Q189 B
 Q190 B
 Q191 $x^2 - 2x - 4 = 0$
 Q192 C
 Q193 $x^2 - 7x + 9 = 0$
 Q194 $x^2 + 10x + 49 = 0$
 Q195 D
 Q196 C
 Q197 $4x^2 + 3x + 2 = 0$
 Q198 A
 Q199 $x^2 + 8x + 24 = 0$
 Q200 D
 Q201 $-2 < m < -1$
 Q202 $-1 < m < -\frac{1}{2}$
 Q203 A
 Q204 B
 Q205 $-6 < m < -2$
 Q206 $k > 10$
 Q207 C
 Q208 B
 Q209 $m > -4$
 Q210 $k > -3$
 Q211 A
 Q212 D
 Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary
 Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary
 Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary
 Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c ; other roots a and b .Q221 $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$ Q222 For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$ Q223 $-4 < m < -3$ Q224 $m > 5$ Q225 $m < -4$ Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$ Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$